

Un-Human

A LITTLE OVER 10 YEARS AGO (JUNE 2011), IN THIS SAME COLUMN I EXPRESSED AN OPINION THAT LOOKED AT OUR QUEST FOR GENERAL AI. In that column I wrote out my thought process for why AI needs to be more human (or at least appear to be). And the trait I felt AI was missing was the ability to gain our trust, which in some rare cases, might even call for an AI to be *deceitful*.

Imagine looking into a smart mirror and asking it, "Do I look good today?" It has two options. The first is to tell you something 100% factually correct, "We have a database of 42,365,278 smart mirror users, and using the rules of attractiveness from social media reactions, you are uglier than 97.342% of all smart mirror users." The second option is to say, "You're looking hot today <user>". Which option do you think will cause smart mirrors to sell more units?

A decade ago I thought that in order to be functional and learn, AI absolutely had to tell white lies, be diplomatic, gain our trust just as a human would, and not be 100% factual all the time... in 2021 I find myself thinking it's totally the wrong way to go about it.

My major concern is that we might try and design our robots (or AI) to be *too human*. And what does it mean to be human? We like to think of scientists like Einstein, inventors like Tesla, physical marvels like Usain Bolt, etc., as examples of the human condition. However, that's just us lying to ourselves. We're petty, mean, selfish, greedy, have serious problems with addiction, and rarely ever change – even when our bad habits are literally killing us! The person who cuts you off in traffic, the scammer trying to get you to share your credit card information, the pickpockets, rapists, murderers and the like are the other end of the human range, and they outnumber the Einsteins of the world a million to one!

Besides, technology (and AI as a result) is not built for the Einsteins of the world, and to be fair, neither is it built for the murderers of the world. Instead, it is built for the average human, and boy is the average human flawed. We waste way too much time on stupid social media videos, we live based on a system of rewarding ourselves with things that make us feel good in the moment – drugs, alcohol, smoking, sex, dopamine hits, adrenaline thrills, and more!

How many of us have sat binge-watching a show instead of sleeping on time? How often do we procrastinate at work? Gorge ourselves on unhealthy food, or cheat and lie to the people we love? Chances are, building an AI to please the average human will build in preset biases into the system. And in a learning system, this manifests as a problem pretty quickly.

This is because we're trying to get AI to learn, and move away from hard coded programming. An integral part of "learning" is the process of creating the "desire" to actually learn. With humans, kids specifically, we use reward or punishment to push them in a direction of wanting to learn. It also helps that the human condition is usually geared towards being inquisitive.

One such experiment run by some researchers at OpenAI used a boat racing game called CoastRunners. You can read their blog post at <https://digit.in/3opqXQx> which explains why almost all humans would immediately understand that the goal of the game was to finish faster than opponents, and that the game was actually a race. The AI, however, was given an incentive of chasing a high score, and not the fastest time. This means that the dopamine hit for the AI happens when it increases its score, and the hit is bigger with a bigger jump in the score.

As it turns out, CoastRunners also rewards people for collecting power-ups and not just finishing the fastest. In fact, collecting as many power-ups as it could would actually give it a higher score. The result? An AI that drives the boat in tight circles so as to pick up as many power-ups as possible as soon as they spawn back into existence!

That infinite loop of chasing power-ups that the AI was stuck in could also be construed as a robot addiction. After all, seeking very short term rewards are exactly how addicts behave as they spiral out of control.

Almost all machine-learning algorithms are being designed around the reward and punishment ethos, just as we raise our young. And that is a scary thought... If things continue along the same path, I wouldn't be surprised if in a few decades AIs need mental help counselling! I just hope we come to our senses and make AI more un-human. **d**



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